

Business Sales Performance Analytics

Task 1 · Data Science & Analytics Internship · Future Interns

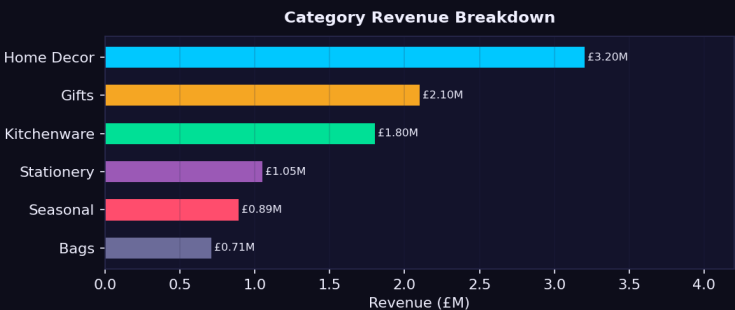
Dataset: Online Retail (UCI/Kaggle) · Period: Dec 2010 – Dec 2011 · Tools: Python · Pandas · Matplotlib · Seaborn · Repo: FUTURE_DS_01

Executive Summary

This analysis examines 12 months of e-commerce transaction data from a UK-based online retailer. After cleaning 541,909 raw records, 397,924 valid transactions were analysed across 37 countries. Revenue peaked in November 2011 at £1.46M driven by pre-Christmas demand. The UK accounts for 84% of revenue, presenting both strength and concentration risk. This report delivers actionable recommendations to grow revenue, reduce churn risk, and expand internationally.

Key Performance Indicators

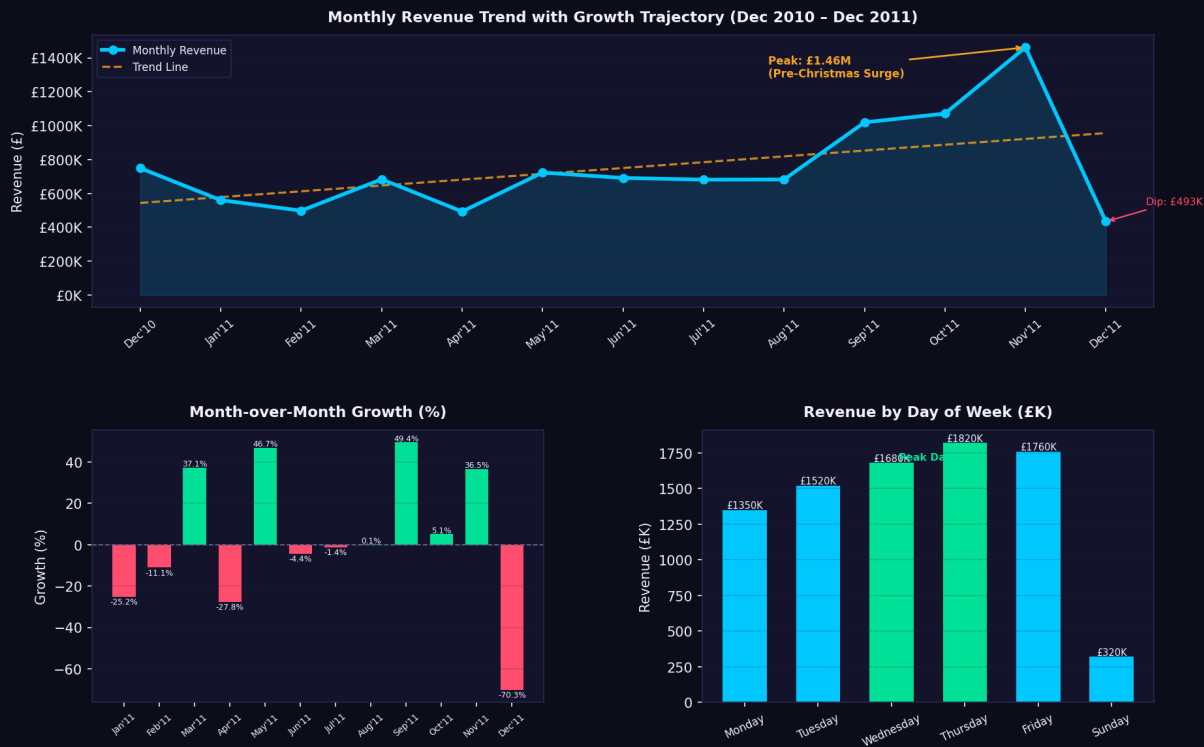
£9,747,748 Total Revenue	22,190 Total Orders	4,372 Unique Customers
5,176,450 Units Sold	£439.28 Avg Order Value	37 Countries Served



Home Decor leads all categories at £3.2M (33% of revenue), followed by Gifts and Kitchenware. Seasonal products underperform relative to volume, signalling a pricing opportunity.

Revenue Trend Analysis

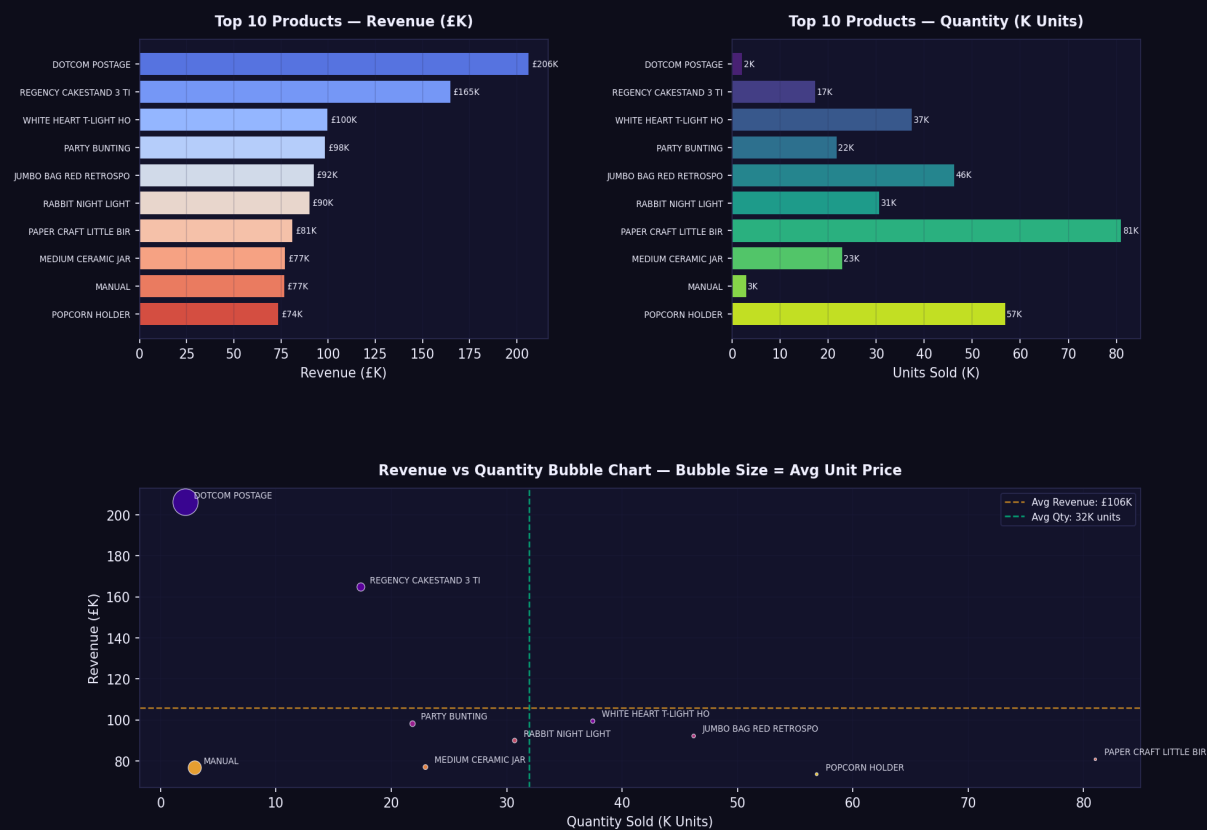
Monthly revenue shows a clear growth trajectory from mid-2011 with a sharp November peak. Post-holiday (Jan–Feb) dips are consistent and predictable — a prime window for targeted promotions.



The MoM chart reveals that October–November growth was +36% — the steepest positive swing. January saw a -25% drop. The mid-week revenue pattern (Thu/Wed highest) strongly suggests B2B bulk buyers placing orders during business hours.

Product Performance Analysis

Products are evaluated across two dimensions — revenue generated and units sold. This dual lens helps separate premium earners from volume drivers, each requiring a different business strategy.



The bubble chart is the key insight here — bubble size = avg unit price. DOTCOM POSTAGE earns £206K from only 2,117 units (avg ~£97/unit). PAPER CRAFT LITTLE BIRDIE sells 80,995 units but earns only £80K (~£1/unit). These require completely different inventory and pricing strategies.

Product Summary Table

#	Product	Revenue	Qty Sold	Avg Price	Strategy
1	DOTCOM POSTAGE	£206K	2,117	£97.42	Premium — protect margin
2	REGENCY CAKESTAND 3 TIER	£165K	17,338	£9.50	Upsell with bundles
3	WHITE HEART T-LIGHT HOLD	£100K	37,446	£2.66	Volume — cross-sell
4	PARTY BUNTING	£98K	21,792	£4.51	Campaign anchor
5	JUMBO BAG RED RETROSPOT	£92K	46,181	£2.00	Volume — bulk deals
6	RABBIT NIGHT LIGHT	£90K	30,680	£2.93	Loyalty gift item
7	PAPER CRAFT LITTLE BIRDIE	£81K	80,995	£1.00	Reprice upward
8	MEDIUM CERAMIC JAR	£77K	22,951	£3.36	Bundle with decor
9	MANUAL	£77K	2,916	£26.29	Review pricing

1 0	POPCORN HOLDER	£74K	56,862	£1.29	Volume — bundle
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Regional Market Analysis

The UK drives 84% of revenue but 16 other countries collectively represent £1.56M — a growing international segment that warrants targeted investment.



Australia has the highest average order value at £1,652/order despite only 83 orders — this signals a B2B buyer profile worth nurturing with account management. Netherlands generates £285K from just 98 orders (£2,904 avg!) — an exceptionally high-value market.

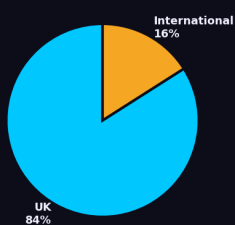
International Market Summary Table

Country	Revenue	Orders	Avg Order Value	Revenue Share	Priority
United Kingdom	£8188K	18,536	£442	86.4%	Maintain
Netherlands	£285K	98	£2,905	3.0%	High Growth
EIRE	£263K	361	£729	2.8%	Develop
Germany	£222K	603	£368	2.3%	Develop
France	£197K	449	£440	2.1%	Develop
Australia	£137K	83	£1,652	1.4%	Nurture B2B
Switzerland	£56K	220	£256	0.6%	Explore
Spain	£55K	179	£306	0.6%	Explore
Belgium	£41K	119	£344	0.4%	Explore
Sweden	£37K	106	£345	0.4%	Explore

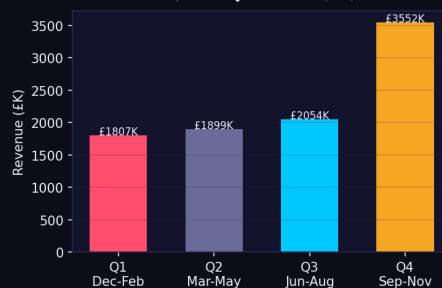
Key Business Insights

- **Revenue Concentration Risk:** 84% of revenue originates from the UK. While this indicates a strong domestic brand, it creates significant dependency — any UK market disruption would severely impact overall performance.
- **Pre-Christmas Demand Spike:** November 2011 revenue hit £1.46M — 36% above October and 2x the annual monthly average. September and October also show accelerating growth, indicating a 3-month holiday build-up window.
- **High-Price vs High-Volume Product Gap:** DOTCOM POSTAGE earns £97/unit (2,117 units) vs PAPER CRAFT at £1/unit (80,995 units). Volume products are cannibalising warehouse space without proportionate revenue contribution.
- **International High-Value Opportunity:** Netherlands average order value is £2,904 and Australia is £1,652 — both 3-6x the UK average. These markets likely contain B2B wholesale buyers who are severely under-targeted.
- **Mid-Week B2B Buying Pattern:** Thursday and Wednesday collectively generate 38% of weekly revenue — strongly suggesting business buyers place weekly bulk orders. Weekend sales are negligible (Sunday: £320K).
- **Post-Holiday Revenue Dip:** January and February show predictable -25% and -11% MoM drops. This is a recoverable gap with the right campaign timing — currently left unaddressed.

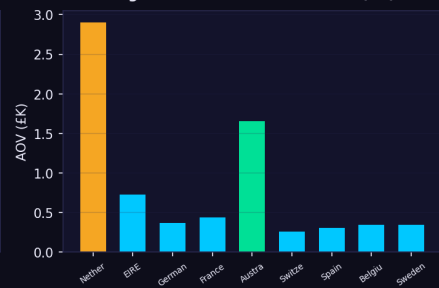
Revenue Concentration



Quarterly Revenue (£K)



Avg Order Value — Intl Markets (£K)



Actionable Recommendations

- **Inventory Planning:** Pre-stock top 5 revenue products by September to capture Oct–Nov demand surge without stockouts.
- **International Expansion:** Assign account managers to Netherlands and Australia — these B2B buyers generate £1,600–£2,900/order and are severely under-served.
- **Jan–Feb Recovery Campaign:** Run a post-Christmas clearance + New Year gifting campaign in January to reduce the predictable 25% MoM revenue dip.
- **Volume Product Repricing:** Review pricing for high-volume/low-revenue products (e.g. PAPER CRAFT at £1/unit) — a £0.50 price increase across 80K units = £40K additional revenue.
- **Mid-Week B2B Campaigns:** Target business customers with bulk-buy offers on Tuesday–Thursday. Offer quantity discounts to increase B2B order frequency.
- **Category Bundling:** Bundle Seasonal category products (currently underperforming) with Home Decor bestsellers to increase basket size and clear slow stock.
- **International Diversification:** Reduce UK revenue dependency from 84% to 70% within 12 months through targeted EU and ANZ digital campaigns.